

Python Programming

Unit 06 – Lecture 05 Notes

Matplotlib Fundamentals (pyplot, labels, grid)

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1 Lecture Overview

Matplotlib is the most commonly used plotting library in Python. This lecture focuses on basic plotting with `matplotlib.pyplot`:

- line plots,
- labels and title,
- grid and legend,
- saving plots to files.

2 Setup (If Needed)

If Matplotlib is not installed:

```
pip install matplotlib
```

3 Core Concepts

3.1 Basic Plot Workflow

Typical steps:

1. prepare x and y data,
2. call `plt.plot(...)` or another plotting function,
3. add labels/title/legend,
4. show or save the plot.

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4]
y = [2, 4, 6, 8]

plt.plot(x, y)
plt.show()
```

3.2 Labels, Title, Grid, Legend

```
plt.plot(x, y, label="y = 2x")
plt.title("Line Plot")
plt.xlabel("x")
plt.ylabel("y")
plt.grid(True)
plt.legend()
```

3.3 Customization

You can customize style:

- color (`color="red"`)
- line style (`linestyle="-"`)
- marker (`marker="o"`)

3.4 Saving Plots

```
plt.savefig("images/line_plot.png", dpi=150)
```

Tip: create the output folder before saving.

4 Demo Walkthrough

File: demo/matplotlib_line_plot_demo.py

This demo creates a customized plot and saves it into images/.

5 Interactive Checkpoints (with Solutions)

Checkpoint 1 Solution

Question: What does `plt.grid(True)` do?

Answer: It draws a grid on the plot background to improve readability.

Checkpoint 2 Solution

Question: Why labels and titles?

Answer: They explain what the plot represents; without labels, a plot can be confusing or misleading.

6 Practice Exercises (with Solutions)

Exercise 1: Plot Squares

Task: Plot $y = x^2$ for $x = 1..10$.

Solution:

```
import matplotlib.pyplot as plt

x = list(range(1, 11))
y = [i*i for i in x]

plt.plot(x, y, marker="o")
plt.title("y = x^2")
plt.xlabel("x")
plt.ylabel("y")
plt.grid(True)
plt.show()
```

7 Exit Question (with Solution)

Question: function used to save a plot?

Answer: `savefig`

Appendix: Slide Deck Content (Reference)

The material below is a reference copy of the slide deck content. Exercise solutions are explained in the main notes where applicable.

Title Slide

Repository: <https://github.com/tali7c/Python-Programming>

Quick Links

[Core Concepts](#) [Demo](#) [Interactive](#) [Summary](#)

Agenda

- Core Concepts
- Demo
- Interactive
- Summary

Learning Outcomes

- Create basic plots using Matplotlib pyplot
- Add titles, labels, legends, and grids
- Customize markers, line style, and colors
- Save plots to image files

Matplotlib and pyplot

- Matplotlib is a plotting library for Python
- `matplotlib.pyplot` provides a simple plotting API

Basic Line Plot

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4]
y = [2, 4, 6, 8]

plt.plot(x, y)
plt.show()
```

Labels, Title, Grid, Legend

```
plt.plot(x, y, label="y = 2x")
plt.title("Line Plot")
plt.xlabel("x")
plt.ylabel("y")
plt.grid(True)
plt.legend()
```

Saving a Plot

```
plt.savefig("images/line_plot.png", dpi=150)
```

Demo: Customized Line Plot

- File: demo/matplotlib_line_plot_demo.py
- Saves a plot image to images/

Checkpoint 1

Question: What does `plt.grid(True)` do?

Checkpoint 2

Question: Why do we add labels and titles to plots?

Think-Pair-Share

Discuss:

- What makes a plot misleading? Give 2 examples.

Key Takeaways

- Use pyplot for quick plotting
- Always label axes and add a title
- Grids and legends improve readability
- Save plots using `savefig`

Exit Question

Write the function name used to save a plot to a file.